

IP Addressing Plan

Overview

This document describes the IPv4 addressing scheme used throughout the Enterprise Network Infrastructure & Security Lab.

The network uses a combination of /24, /28, /30, and /32 networks to support user VLANs, management networks, point-to-point routed links, and router loopback interfaces.

VLAN Addressing

VLAN	Purpose	Network	Subnet Mask	Addressing
VLAN 10	PCs	10.1.0.0/24	255.255.255.0	User Devices
VLAN 20	Phones	10.2.0.0/24	255.255.255.0	Voice Devices
VLAN 30	Servers	10.3.0.0/24	255.255.255.0	Server Infrastructure
VLAN 40	Wi-Fi	10.6.0.0/24	255.255.255.0	Wireless Clients
VLAN 99	Management	10.0.0.0/28	255.255.255.240	Network Management

VLANs were used to logically separate different categories of network traffic. The lab configuration confirms VLANs for PCs, phones, servers/Wi-Fi, and management depending on the network segment.

Core Switch SVI Addressing

DSW-A1

Interface	IP Address	Network	Purpose
VLAN 10	10.1.0.2	10.1.0.0/24	PCs
VLAN 20	10.2.0.2	10.2.0.0/24	Phones
VLAN 40	10.6.0.2	10.6.0.0/24	Wi-Fi
VLAN 99	10.0.0.2	10.0.0.0/28	Management
Loopback0	10.0.0.79/32	10.0.0.79/32	Router ID/Loopback

DSW-A1 also uses routed interfaces 10.0.0.46 and 10.0.0.62 for its point-to-point connections.

DSW-A2

Interface	IP Address	Network	Purpose
VLAN 10	10.1.0.3	10.1.0.0/24	PCs
VLAN 20	10.2.0.3	10.2.0.0/24	Phones
VLAN 40	10.6.0.3	10.6.0.0/24	Wi-Fi
VLAN 99	10.0.0.3	10.0.0.0/28	Management
Loopback0	10.0.0.80/32	10.0.0.80/32	Router ID/Loopback

DSW-A2 uses 10.0.0.50 and 10.0.0.66 on its routed interfaces.

DSW-B1

Interface	IP Address	Network	Purpose
VLAN 10	10.3.0.2	10.3.0.0/24	PCs
VLAN 20	10.4.0.2	10.4.0.0/24	Phones
VLAN 30	10.5.0.2	10.5.0.0/24	Servers
VLAN 99	10.0.0.18	10.0.0.16/28	Management
Loopback0	10.0.0.81/32	10.0.0.81/32	Router ID/Loopback

DSW-B1 uses 10.0.0.54 and 10.0.0.70 on its routed interfaces.

DSW-B2

Interface	IP Address	Network	Purpose
VLAN 10	10.3.0.3	10.3.0.0/24	PCs
VLAN 20	10.4.0.3	10.4.0.0/24	Phones
VLAN 30	10.5.0.3	10.5.0.0/24	Servers
VLAN 99	10.0.0.19	10.0.0.16/28	Management
Loopback0	10.0.0.82/32	10.0.0.82/32	Router ID/Loopback

The routing table confirms 10.3.0.0/24, 10.4.0.0/24, and 10.5.0.0/24 as the VLAN networks on the B-side of the topology.

Core Switching Infrastructure

CSW1

Interface	IP Address	Network	Purpose
GigabitEthernet1/0/1	10.0.0.34	10.0.0.32/30	Routed Link
Port-channel1	10.0.0.41	10.0.0.40/30	EtherChannel/Routed Link
Gi1/1/1	10.0.0.45	10.0.0.44/30	Routed Link
Gi1/1/2	10.0.0.49	10.0.0.48/30	Routed Link
Gi1/1/3	10.0.0.53	10.0.0.52/30	Routed Link
Gi1/1/4	10.0.0.57	10.0.0.56/30	Routed Link
Loopback0	10.0.0.77/32	/32	Router ID/Loopback

These addresses are confirmed by the CSW1 interface output.

CSW2

Interface	IP Address	Network	Purpose
GigabitEthernet1/0/1	10.0.0.38	10.0.0.36/30	Routed Link
Port-channel1	10.0.0.42	10.0.0.40/30	EtherChannel/Routed Link
Gi1/1/1	10.0.0.61	10.0.0.60/30	Routed Link
Gi1/1/2	10.0.0.65	10.0.0.64/30	Routed Link
Gi1/1/3	10.0.0.69	10.0.0.68/30	Routed Link
Gi1/1/4	10.0.0.73	10.0.0.72/30	Routed Link
Loopback0	10.0.0.78/32	/32	Router ID/Loopback

CSW2's interface and routing information confirms these point-to-point networks and its 10.0.0.78 loopback.

Router Addressing

R1

Interface	IP Address	Network	Purpose
GigabitEthernet0/0	10.0.0.33	10.0.0.32/30	Internal Routing
GigabitEthernet0/1	10.0.0.37	10.0.0.36/30	Internal Routing
GigabitEthernet0/1/0	203.0.113.6	203.0.113.4/30	External/WAN
Loopback0	10.0.0.76/32	/32	Router ID/Loopback

R1 has a static default route through 203.0.113.5, while its internal routes are learned through OSPF.

Point-to-Point Networks

The topology uses /30 networks for routed point-to-point links.

Network	Usable Addresses	Purpose
10.0.0.32/30	10.0.0.33–10.0.0.34	R1 ↔ CSW1
10.0.0.36/30	10.0.0.37–10.0.0.38	R1 ↔ CSW2
10.0.0.40/30	10.0.0.41–10.0.0.42	CSW1 ↔ CSW2
10.0.0.44/30	10.0.0.45–10.0.0.46	CSW1 ↔ DSW-A1
10.0.0.48/30	10.0.0.49–10.0.0.50	CSW1 ↔ DSW-A2
10.0.0.52/30	10.0.0.53–10.0.0.54	CSW1 ↔ DSW-B1
10.0.0.56/30	10.0.0.57–10.0.0.58	CSW1 ↔ DSW-B2
10.0.0.60/30	10.0.0.61–10.0.0.62	CSW2 ↔ DSW-A1
10.0.0.64/30	10.0.0.65–10.0.0.66	CSW2 ↔ DSW-A2
10.0.0.68/30	10.0.0.69–10.0.0.70	CSW2 ↔ DSW-B1
10.0.0.72/30	10.0.0.73–10.0.0.74	CSW2 ↔ DSW-B2

The routing tables confirm these /30 networks as directly connected or learned through OSPF, demonstrating redundant Layer 3 paths between the network devices.

Loopback Addressing

Device	Loopback Address	Purpose
R1	10.0.0.76/32	Router ID
CSW1	10.0.0.77/32	Router ID
CSW2	10.0.0.78/32	Router ID
DSW-A1	10.0.0.79/32	Router ID
DSW-A2	10.0.0.80/32	Router ID
DSW-B1	10.0.0.81/32	Router ID
DSW-B2	10.0.0.82/32	Router ID

The routing tables show these /32 loopback addresses being advertised and learned through OSPF.

External/WAN Addressing

The external network uses:

203.0.113.4/30

Device	Interface	IP Address
R1	GigabitEthernet0/1/0	203.0.113.6
Upstream Gateway	—	203.0.113.5

R1 uses 203.0.113.5 as its default gateway.

Routing Design

OSPF is used as the primary dynamic routing protocol for the internal network.

The routing tables demonstrate OSPF-learned routes using the **O** designation and redundant next-hop paths. For example, internal VLAN networks such as 10.1.0.0/24 through 10.6.0.0/24 are learned through multiple paths in the topology.

R1 provides the external default route using a static route:

0.0.0.0/0 → 203.0.113.5

This provides a path from the internal OSPF domain toward the external network.

Addressing Design Summary

The network uses:

- **/24 networks** for user and service VLANs
- **/28 networks** for management segments
- **/30 networks** for point-to-point routed connections
- **/32 networks** for loopback interfaces
- **203.0.113.4/30** for the external/WAN connection